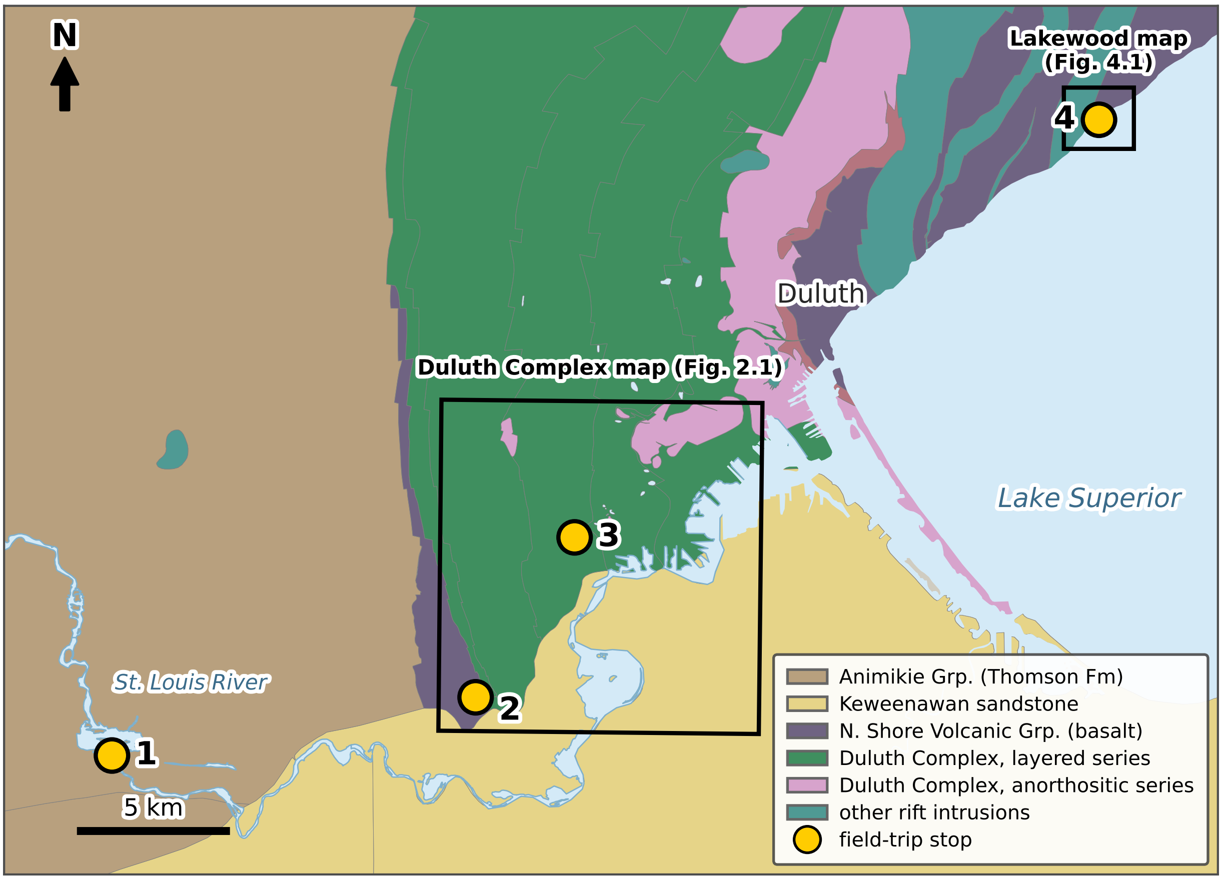
The Duluth Complex: From Footwall to Flows

2026 IRM Summer School Field Trip

25 July 2026



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Colophon
This guide was typeset with LaTeX using the kaobook class (https://github.com/fmarotta/kaobook).

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Introduction

Welcome to the field trip of the 2026 IRM Summer School. This is a one-day excursion from the Twin Cities up to the Duluth area, where in a single day we can stand on rocks that record two of the defining events in the geologic history of this part of Laurentia, separated from one another by 700 million years: the ca. 1.8 Ga Paleoproterozoic Penokean orogeny, and the ca. 1.1 Ga Midcontinent Rift.

The day follows an arc built around the base of the Duluth Complex. At **Thomson Dam** we begin in the metasedimentary footwall: deformed turbidites of the Paleoproterozoic Thomson Formation, cut by diabase dikes emplaced during the early, reversed-polarity stage of Midcontinent Rift magmatism. These are the kind of sulfur-bearing country rocks thought to have sourced sulfur to the magmatic sulfide deposits of the Duluth Complex, and the dikes would have fed magma to rift lavas at the surface. At the **DWP Trail** we meet those lavas (the Ely's Peak basalts, the earliest eruptions of rift volcanism) where they form the immediate, thermally metamorphosed footwall to the Duluth Complex when it intruded. We will walk an old railway grade across the basal contact and up into the layered series itself. Along **Skyline Parkway** we climb structurally higher still, through the cyclic zone of the layered series and into the anorthositic series that overlies it. A final stop along the shore at **Lakewood** brings us to rift lava flows — the surface expression of the Duluth Complex magmatic system.

Itinerary overview

Stop 1 — Thomson Dam: deformation, dikes, and a scavenger hunt in the Thomson Formation (Chapter 1).
Stop 2 — The DWP Trail: Ely's Peak basalts and the base of the Duluth Complex (Chapter 2).
Stop 3 — Skyline Parkway: the layered and anorthositic series of the Duluth Complex; one to three sub-stops as time allows (Chapter 3).
Stop 4 — Lakewood: pahoehoe lava flows of the North Shore Volcanic Group along the shore (Chapter 4).

Thomson Dam — deformation, dikes, and a scavenger hunt

1

Our first stop is the type locality of the Thomson Formation, exposed along the St. Louis River at Thomson Dam. In a single outcrop belt it records both of the events that frame this trip: Paleoproterozoic folds and cleavage in the Thomson Formation, cut by diabase dikes emplaced during the early, reversed-polarity stage of Midcontinent Rift magmatism. Rather than a formal set of show-and-tell stops, we will spend our hour here on a scavenger hunt — looking closely, in small groups, for the features that tell this two-part story.

1.1 Location and logistics

Location: 46.66491°N, 92.40379°W (Thomson Dam, Carlton County; Cloquet quadrangle). Park at the pull-out next to the Highway 210 bridge over the St. Louis River — point 1 on the base map (Fig. 1.1).

Facilities: there is a porta-potty at the parking lot (46.665759°N, 92.40334°W).

Time: one hour for the scavenger hunt.

Safety

The exposures are along the St. Louis River below a dam, and the rock can be slick where wet. Be careful with your footing as you move between outcrops.

1.2 Motivating questions

Structural considerations

As you go forth on the scavenger hunt, keep these two questions in mind.

- **Is the Thomson Formation flat-lying or deformed?** What do the bedding attitudes, the folds, and the cleavage tell you about the strain these rocks have taken?
- **Are the Midcontinent Rift Carlton dikes vertical or tilted?** And what does their orientation — together with the fact that they cut the cleavage without being folded — tell you about what happened to this crust *after* the dikes were emplaced?

The second question matters directly for paleomagnetism: a dike's remanence records the field direction at the time it cooled, so whether the dike (and the crust it sits in) has since been tilted determines whether that direction can be read as originally emplaced or needs a structural correction.

1.3 The scavenger hunt

Scavenger hunt — one hour

In small groups, see how many of the following you can find. Photograph each one, and note where you found it on the base map (Fig. 1.1). We will regroup at the end to compare finds and to discuss the two questions above.

Midcontinent Rift dikes

- ▢ Basaltic dike less than 1 m wide
- ▢ Basaltic dike more than 2 m wide
- ▢ Columnar jointing in a dike

Veins

- ▢ Calcite vein
- ▢ Quartz vein

Sedimentary and structural features of the Thomson Formation

- ▢ Current ripples
- ▢ Planar lamination
- ▢ Graded bedding / grain-size variation
- ▢ Diagenetic concretions
- ▢ Cleavage plane
- ▢ Kink bands *(narrow, sharply bounded bands across which the cleavage is abruptly bent to a new orientation)*
- ▢ Twist hackles *(en-echelon steps where a fracture surface breaks into offset segments as the crack front twists)*
- ▢ Anticline

Edible extras

- ▢ Raspberries
- ▢ Blueberries

1.3.1 A few hints

The **columnar jointing** in the dikes is a way into the second question above. Where a chilled margin or contact is exposed, you can measure the attitude of the dike wall directly. Where it is not, the columns give you an indirect insight: columns grow perpendicular to the cooling surface, so in a sheet-like intrusion they form normal to the walls. So when you find a dike, look at how its columns are oriented: are they horizontal, vertical, or somewhere in between? If the columns are horizontal, what does that tell you about the attitude of the dike walls — and if they are inclined instead? Do the two lines of evidence — the wall you can measure and the columns you can only infer from — agree? Once you have reasoned back to the orientation of the dike, ask the follow-up: is that the orientation the dike had when it cooled, or has something tilted it since?

Figure 1.1: Base map for the Thomson Dam scavenger hunt, built on public-domain USDA/USGS NAIP aerial imagery. North is up: the dam is at the top, the Highway 210 bridge and parking area across the middle, and the pedestrian/railroad bridge at the bottom. Numbered points — (1) parking and porta-potty; (2) a good place to start the hunt, on the rock exposures below the dam; (3) a way down to the rocks, watching your footing on the rip-rap boulders; (4) the pedestrian path; (5) the river overview where we regroup after an hour. A nice walkable path runs between points 4 and 5, with Thomson Formation exposed along it. Mark your finds as you go.

1.4 Geologic setting

The Thomson Formation is part of the Animikie Group, which accumulated in a foreland basin associated with the Paleoproterozoic Penokean orogeny ca. 1.85 billion years ago. It consists of graywacke, siltstone, and slate deposited by turbidity currents (Morey and Ojakangas, 1970), and is temporally equivalent to the Virginia Formation of the Mesabi Iron Range. This is the type locality: roughly 200 m of strata are exposed between the dam north of Highway 210 and the railroad bridge to the south, in approximately equal proportions of graywacke, siltstone, and slate, metamorphosed to lower greenschist facies.

Unlike the gently dipping, essentially undeformed Animikie strata of the Mesabi Range, the Thomson Formation here is folded at a range of scales and carries a well-developed, steeply dipping cleavage in its slaty beds; concretions and mud chips are flattened into the plane of that cleavage, and it is locally deformed by kink bands (Holst, 1984; Ojakangas et al., 2011). Graded bedding, cross-bedding, ripple lamination, sole marks, and flame structures record the turbidite origin of the beds and give way-up control (Morey and Ojakangas, 1970). Carbonate-rich concretions are abundant and can be a useful marker of bedding in otherwise massive graywacke.

Cutting across all of this are diabase dikes of the Carlton County swarm (Boerboom, 2009). They are dark gray, fine-grained, granular to ophitic tholeiitic basalt; they strike about N. 30° E., dip near vertical, have chilled margins, and display columnar jointing (Boerboom, 2009). Mapped dikes range from about 1 to 18 m thick, though thinner ones occur as well. The swarm carries a *reversed-polarity* magnetization with directions consistent with the early ca. 1108–1105 Ma magmatic pulse of the Midcontinent Rift (Green et al., 1987); every dike sampled in the county is reversed except two (Boerboom, 2009). The North Shore Volcanic Group crops out ~13 km to the east, and these dikes may have fed those lavas (Ojakangas et al., 2011). Because the dikes are undeformed where they cut the folded and cleaved metasedimentary rocks, they place a hard younger bound on the age of the deformation.

1.5 An IRM connection: the magnetic fabric of these rocks

In the early 1990s, two studies out of the Institute for Rock Magnetism used magnetic-fabric methods to gain insight into the strain recorded in the Thomson Formation (Johns et al., 1992; Sun et al., 1995), and the distinction they drew is one worth carrying into the field with a rock-magnetist's eye.

Johns et al. (1992) measured the anisotropy of low-field magnetic susceptibility (AMS) and of anhysteretic susceptibility (AAS) across the formation. The AAS is essentially the AARM discussed below — the anisotropy of anhysteretic remanence, isolating the magnetite grains — expressed in susceptibility units (ARM divided by the bias field) so it can be compared side by side with the AMS. In the northern zone — the zone we are standing in at Thomson Dam — the susceptibility

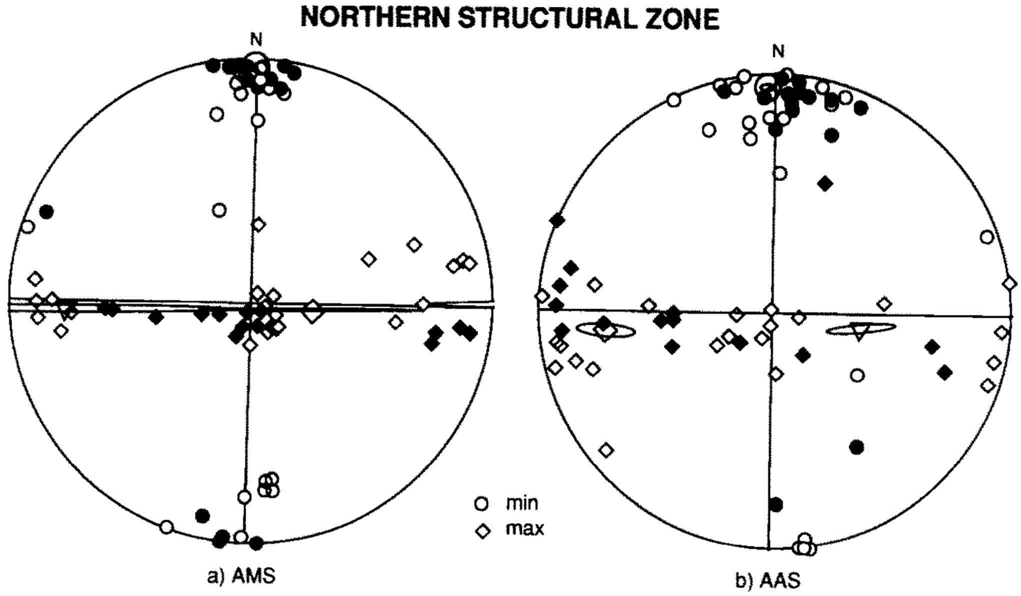

Figure 1.2: Magnetic-ellipsoid axes for the northern structural zone — the zone exposed here at Thomson Dam — from Johns et al. (1992). Equal-area, lower-hemisphere stereonets of (a) the low-field magnetic susceptibility (AMS) and (b) the anhysteretic susceptibility (AAS); circles are minimum axes, diamonds maximum axes, and small ellipses are the 95% confidence regions on the matrix-mean axes. The northern-zone cleavage is upright and strikes roughly east–west, so it projects as the vertical (north–south) plane whose pole is horizontal and points north–south — plotting at the north and south edges of the net. In both panels the minimum axes cluster tightly there, at the north and south of the primitive: they are normal to the steep cleavage. The maximum axes, by contrast, do not form a tight cluster but smear along the east–west diameter, spreading *within* the cleavage plane — the signature of an oblate, flattening fabric with no strong lineation. The low-field magnetic fabric thus faithfully tracks the tectonic fabric. Filled symbols are fine-grained samples (slate and finely laminated graywacke); open symbols are coarse-grained samples.

ellipsoids are oblate, plotting in the flattening field of a Flinn diagram with minimum axes normal to the cleavage (Fig. 1.2): the magnetic fabric tracks the tectonic fabric. But the AMS is a *mixture*. Separating the paramagnetic contribution (carried by chlorite) from the ferrimagnetic one (carried by magnetite) showed that the low-field susceptibility fabric is compositionally controlled, and that the two mineral subfabrics need not record the same thing.

Sun et al. (1995) sharpened the point in a way close to the heart of this Summer School. The paleomagnetic remanence and the anisotropy of anhysteretic remanent magnetization (AARM) — both carried by the ferrimagnetic grains — record deposition and primary compaction; "neither NRM nor AARM were much affected by tectonic deformation." The low-field AMS, by contrast, is dominated by paramagnetic chlorite and faithfully records the tectonic fabric, with minima normal to the cleavage in both the north and the south. In one and the same rock, then, the remanence fabric and the susceptibility fabric tell different stories — one sedimentary, one tectonic — a clean reminder that *what carries a magnetic fabric determines what that fabric means*. Comparing the AMS to the strain recorded by the crystallographic preferred orientation of chlorite, Sun et al. (1995) went on to raise the possibility that the cleavage we see here in the north formed *earlier* in the deformation history than has usually been assumed — belonging to the belt's first phase of folding rather than a later one. Which episode of deformation this cleavage records is still debated, and it is a nice example of a magnetic-fabric measurement bearing on a structural-geology question.

2 The DWP Trail — Ely's Peak basalts and the base of the Duluth Complex

At Thomson Dam we stood in the metasedimentary footwall of the Duluth Complex. Those older Animikie rocks matter to the magmatic story we turn to now in two ways. First, sulfur-bearing metasedimentary rocks of the Virginia and Thomson Formations are widely held to have been the principal external source of sulfur for the Cu-Ni-(PGE) sulfide deposits that make the base of the Duluth Complex one of the largest magmatic sulfide deposts on Earth: where mafic magma assimilated rocks from the footwall, it became sulfide-saturated. Second, the reversed-polarity Carlton dikes that cut the Thomson Formation belong to the earliest, reversed stage of Midcontinent Rift magmatism and are of the sort that fed rift lavas to the surface.

At this stop we will see both extrusive and intrusive rocks of the Midcontinent Rift. The **Ely's Peak basalts** are the earliest lavas erupted associated with the Midcontinent Rift, and they form the immediate footwall to the

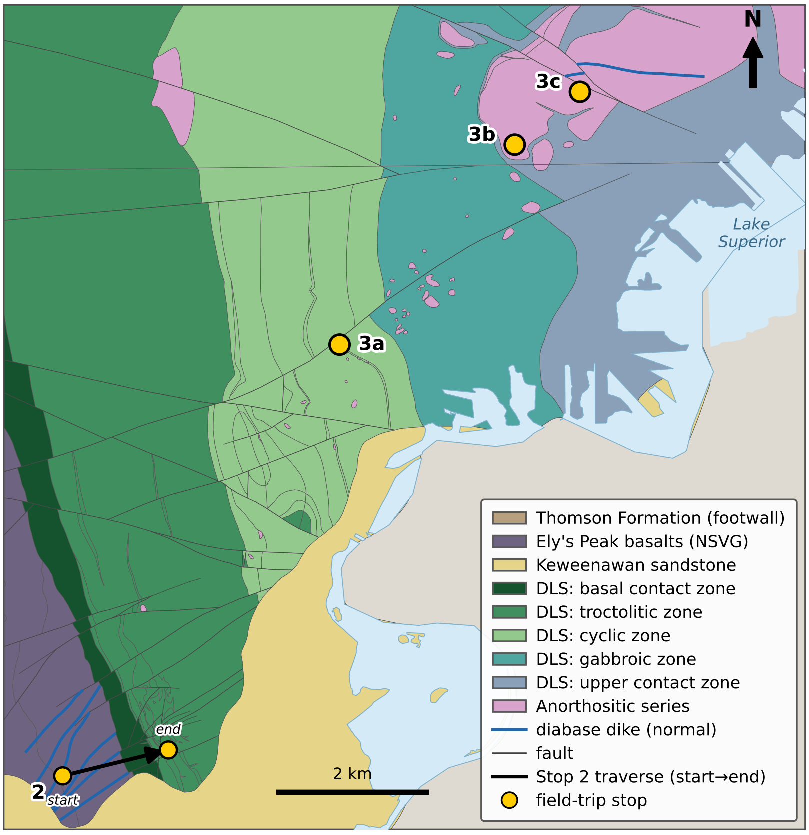

Figure 2.1: Bedrock geologic map of the Duluth Complex from Ely's Peak to Duluth Heights, merged from the Minnesota Geological Survey M-183 (West Duluth) and M-181 (Duluth Heights) coverages (Miller and Green, 2008b; Miller and Green, 2008a), locating the field-trip stops. The layered series forms a northeast-younging succession: from the Ely's Peak basalt footwall and the basal contact zone, up through the troctolitic, cyclic, gabbroic, and upper contact zones, and into the overlying anorthositic series. Stop 2 (the DWP Trail) is drawn as the walked traverse, from a start in the basalt across the basal contact into the troctolitic zone; Stop 3a lies in the cyclic zone at Thompson Hill; Stops 3b and 3c lie in the anorthositic series. Pink bodies within the layered series are inclusions of anorthositic series; blue lines are normal-polarity diabase dikes.

Duluth Complex near Duluth — thermally metamorphosed to hornfels where the gabbro was emplaced against them (Figure 2.1). Walking the former railway grade of the former Duluth, Winnipeg, and Pacific (DWP) Railway that is now a bike/walk path, we pass through those basalts — including a rock tunnel blasted through them — and then cross the basal contact of the Duluth Complex and climb up-section into the layered series. This is Stop 2-1 of Miller (2011), and there are quotes from that guide below.

2.1 Location and logistics

Parking: Short Line Park, 2384 Becks Road, Duluth. There is a porta-potty at the parking lot (46.68011°N, 92.25993°W).

Route: from the lot, take the path up onto the DWP grade and walk to the northeast. The **railroad tunnel** through the south flank of Ely's Peak (46.67841°N, 92.25323°W) lies ∼0.25 mile along, in the Ely's Peak basalts. Continuing along the grade, the **basal contact** of the Duluth Complex is reached at about 46.68377°N, 92.24496°W; the flagged traverse of Miller (2011) runs ∼400 m up-section from there to the layered troctolite at locality I, near 46.68137°N, 92.23558°W — about 1.5 miles' walk from the trailhead (West Duluth 7.5′ quadrangle; NAD 83 UTM start 557740E, 5170300N, end 558460E, 5170040N).

2.2 Geologic setting

The Ely's Peak basalts are a ∼1.5 km thick succession of tholeiitic lavas at the very base of the North Shore Volcanic Group — the earliest volcanic eruptions into the rift, and the same reversed-polarity, early-rift interval represented by the Carlton dikes we saw at Stop 1. Here, they formed the footwall onto which the Duluth Complex was emplaced. The complex baked them to hornfels.

The Duluth Complex rock we cross into here is troctolite and olivine gabbro of the Duluth Layered Series. As we walk, you can be looking for the grain-size contrast to know when we have crossed into the Duluth Complex. The lavas are fine-grained with vesicles/amygdules while the gabbro/troctolite will lack vesicles and be much coarser grained.

The Duluth Layered Series is built of stacked *macrocycles* interpreted to record successive recharges of magma, but these are subtle and are unlikely to be sorted out on a quick walk. One quick point on mineralization: although the Duluth Complex hosts large-scale Cu-Ni-(PGE) sulfide deposits where mafic magma assimilated sulfur-bearing metasedimentary footwall, the basal contact *here* — where the footwall is basalt that lacks sulfer — carries little sulfide.

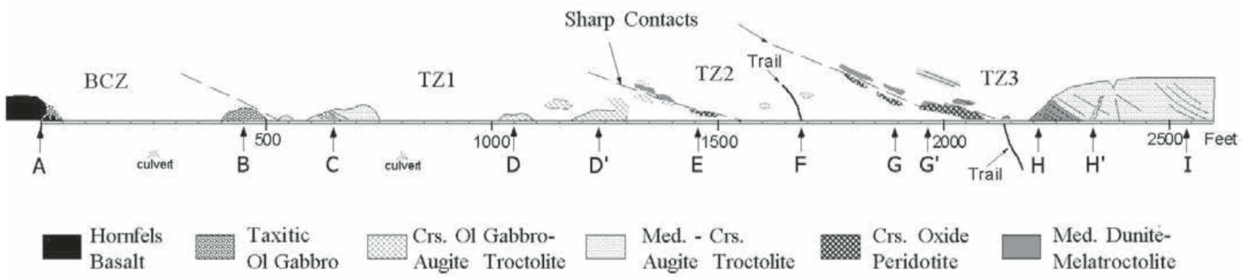

Figure 2.2: Schematic cross-section of the exposures along the north side of the DWP railroad grade, from Miller (2011) (their Fig. 17), showing variability in rock type across the basal contact zone (BCZ) and the lower troctolitic zone (TZ1–TZ3). Lettered flags (A–I) mark localities along Miller's traverse; we walk the full section from the basal contact (A) up through the macrocycles to the layered troctolite at I, described below.

2.3 The approach: Ely's Peak basalts and the tunnel

Before we reach any Duluth Complex rock, the path carries us straight into the Ely's Peak basalts. The **tunnel** — blasted through the south flank of Ely's Peak in 1911 for the DWP Railway — is cut in these lavas, and the rock cuts on either side of it give the first good look at them. Here, a few hundred meters out from the intrusion, the basalt is comparatively fresh: gently dipping (10°–15°) tholeiitic flows, in places showing amygdaloidal flow tops. These are the earliest lavas erupted into the Midcontinent Rift, and they belong to the same early, reversed-polarity interval as the Carlton dikes we examined at Thomson Dam.

The Superior Hiking Trail spur to the summit of Ely's Peak, which climbs off the grade near here, is a fine short hike with panoramic views over the St. Louis River valley — worth a return trip, though it is not on our route today.

What to look for

As we walk the grade, keep answering one question: *are we on basalt, or on gabbro?* Two observations are key:

- ☐ **Vesicles and amygdules** — the cavities left by gas bubbles, either open or filled with secondary minerals. They are a hallmark of lava, so finding them means you are in the Ely's Peak basalt.
- ☐ **Grain size** — the basalt is fine-grained, with crystals too small to pick out by eye; the troctolite and gabbro of the Duluth Complex are coarse-grained, with plagioclase, olivine, and pyroxene crystals easily visible. Even where the basalt is baked to hornfels near the contact, it stays far finer than the cumulate gabbro.

2.4 Walking the section (Stop 2-1 of Miller, 2011)

The descriptions that follow are quoted directly from Stop 2-1 of Miller (2011). Locality letters refer to the flags in Figure 2.2, and the modal classification diagram in the margin (Fig. 2.3) is a good check on the rock names as we go.

> *This stop traverses ~400 m of stratigraphic section of the lower part of the Layered Series at Duluth (DLS). The exposures will be accessed by walking the railroad grade from the western side*

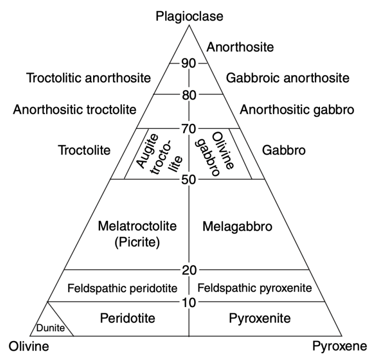

Figure 2.3: Modal classification of coarse-grained plagioclase–olivine–pyroxene cumulates. It does not apply to the fine-grained basalt, but it is a good check on the Duluth Complex rocks: estimate the proportions of plagioclase, olivine, and pyroxene in an outcrop, read off a rock name, and compare it with the names Miller uses in the quoted descriptions below — troctolite, augite troctolite, olivine gabbro, melatroctolite, oxide peridotite, and so on.

of Ely's Peak. Along the way, several extensive exposures of Ely's Peak basalts (including a tunnel) will be encountered. The Ely's Peak basalts form the footwall of the Duluth Complex and show the effects of intense thermal metamorphism by the gabbro. The Ely's Peak basalts are composed of ~1.5 km of gently (10°–15°) dipping, tholeiitic basaltic lavas and represent the earliest volcanic eruptions into the Midcontinent Rift.

Starting at the basal contact, which trends ~N-S in a valley between Ely's and Bardon's peaks, a variety of rock types belonging to the basal contact zone and the lower part of the troctolitic zone of the DLS are exposed in roadcuts and outcrops along the abandoned railroad grade as it arcs around Bardon Peak. Several macrocyclic layers grading in mode and texture are apparent in the lower part of the troctolitic zone and are thought to represent major recharge and differentiation/cooling episodes associated with the early integration stage of DLS emplacement.

Miller's locality descriptions follow — given verbatim in italics, with the intervening flags D–H condensed:

- **A Basal contact** — *The traverse starts at the irregular contact between strongly hornfelsed mafic volcanics of the Ely's Peak basalts and taxitic olivine gabbro and augite troctolite of the basal contact zone (Flag A). Granophyre dikes cut both rock types, but display lobate contacts with the coarse gabbro, suggesting two-liquid mixing. Grout (1918e) cited this mixing of granophyre and gabbro as evidence of silicate immiscibility. An alternative explanation is that the granophyre represents anatectic melts from the footwall, which were not completely assimilated into the mafic magma. Contacts between the basaltic hornfels and the gabbro are commonly characterized by coarse augite prisms oriented perpendicular to the contact.*
- **B Basal contact zone** — *This exposure of taxitic olivine gabbro to augite troctolite with its variable textures (medium-grained to pegmatitic, subophitic to ophitic, nonfoliated to poorly foliated) and variable modal compositions are characteristic of the basal contact zone. In contrast to other layered series intrusions that locally contain abundant Cu-Ni(-PGE) sulfide along their margins, the basal contact zone of the DLS is devoid of significant sulfide mineralization.*
- **C Lower macrocyclic unit of the troctolitic zone** — *Exposed here is a modally layered, medium-grained, moderately foliated, ophitic augite troctolite. This foliated POcf cumulate marks the base of the troctolitic zone and the first of three macrocyclic layers exposed along this traverse. At the east end of this exposure, the rock grades to a medium coarse-grained, augite troctolite to olivine gabbro.*
- **D Upper part of the lower macrocycle** — coarse- to very coarse-grained, non- to poorly foliated, ophitic to subophitic olivine oxide gabbro, capping the lower macrocycle (TZ1).
- **E Contact between the lower and medial macrocycles** — a sharp contact where coarse olivine oxide gabbro passes up into feldspathic, biotitic oxide peridotite and then melatroctolite. The abrupt jump from coarse gabbro to medium-grained melatroctolite is interpreted to mark a major recharge of the magma chamber; the oxide peridotite carries trace sulfide (note local Cu staining) and elevated PGE.

F Middle of the medial macrocycle — medium- to medium coarse-grained, moderately foliated, locally layered augite troctolite (mid-TZ2).

G Contact between the medial and upper macrocycles — massive biotitic oxide peridotite at the top of the medial macrocycle, in abrupt contact with well-foliated feldspathic dunite to melatroctolite; lenticular troctolite layers that thicken westward hint at channel or trough layering, of the kind best displayed just above at I.

H Lower part of the upper macrocycle — well-layered melatroctolite grading up into medium-grained, well-foliated, ophitic augite troctolite (base of TZ3), locally cut by steep bodies of oxide olivine gabbro pegmatite.

I Layered troctolite of the upper macrocycle — the best expression of igneous layering on the stop: *At the highest part of the cut is an exceptional exposure of lenticular layered melatroctolite and leucotroctolite. Similar to the upper exposure at G, trough/channel structures are seemingly implied by the bowing of the layering and the consistent thickening of leucocratic layers in the axis of the trough. Several trough structures have been recognized in the Troctolitic Zone in the Bardon Peak area. The axis of the troughs tend to trend easterly (i.e., in the down-dip direction of regional layering).*

Standing at I, we have walked into the base of the Duluth Complex: from the vesicular Ely's Peak basalt at the tunnel, through hornfels at the contact, and up through macrocycles of troctolite and olivine gabbro to this layered troctolite. From here, we get to journey back through the base and back to the vehicles.

Skyline Parkway — the layered and anorthositic series of the Duluth Complex

3

At the DWP Trail we crossed the base of the Duluth Complex, from the basalt footwall up into the lowest troctolite of the layered series. Skyline Parkway is higher into the complex (Figure 2.1). Roadcuts along the parkway expose the *cyclic zone* — the medial part of the Duluth Layered Series — and, farther on, the anorthositic series that overlies it.

We have three possible stops here, and how many we make will depend on time. Stop 3a, at the Thompson Hill roadcut just below the Minnesota Welcome Center, is a planned stop as we both get to see the Duluth Complex and get a view out to Duluth and Lake Superior. Stops 3b (an anorthosite xenolith) and 3c (the anorthositic series) are ones we will take if time allows. The bedrock map in Figure 2.1 (Chapter 2) shows how these stops step northeastward up through the Duluth Complex, from the base we walked at the DWP Trail into the anorthositic series.

3.1 Stop 3a: the Cyclic Zone of the Layered Series at Thompson Hill

Location: 46.72895°N, 92.20559°W — a roadcut on Skyline Parkway just below the Thompson Hill rest area and Minnesota Welcome Center, at the southwest end of the parkway above West Duluth (West Duluth 7.5′ quadrangle; NAD 83 UTM 560700E, 5175350N). This is Stop 2-5 of Miller (2011).

This 200-m-long roadcut exposes part of the cyclic zone of the Duluth Layered Series (DLS) — structurally well above the basal rocks we saw at the DWP Trail. The cyclic zone is built of stacked *macrocycles*, that are interpreted to record a batch of magma that differentiated in place.

Figure 3.1 shows what Miller (2011) mapped and measured across this cut. Traced from southwest to northeast, the compositions of the cumulus minerals do not simply drift in one direction: olivine (Fo) and pyroxene (En) follow a differentiation trend and then *reverse* at a "cumulus reversal." The rock is dominantly layered oxide gabbro.

The description and interpretation below are quoted from Stop 2-5 of Miller (2011). On the character of the layered gabbro:

> *At another 3 m above this, the coarse gabbro passes into a medium-grained, well-laminated, subpoikilitic olivine-bearing oxide gabbro (samples D and D′) which displays layering of olivine oikocryst concentration and elsewhere isomodal layers rich in Fe-Ti oxide and pyroxene. The very strong foliation and subhedral to euhedral habit of cumulus phases (plagioclase, pyroxene, and ilmenite) impart an adcumulate texture to this rock.*

And on what the cumulus reversal might record :

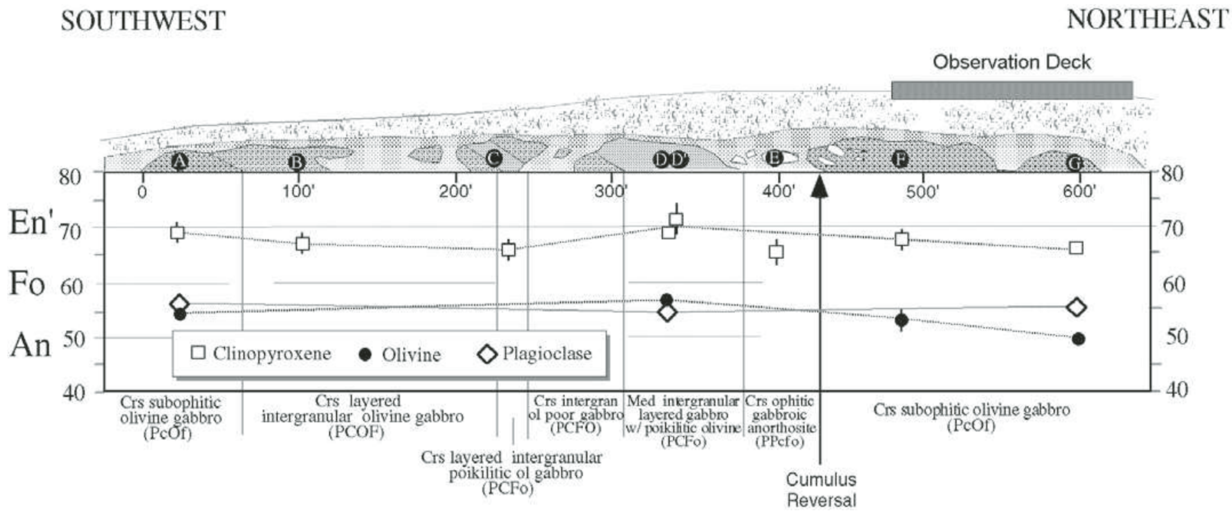

Figure 3.1: Geology and cryptic variation along the Skyline Parkway roadcut at Thompson Hill, from Miller (2011) (their Fig. 20). The view is to the north and the dip of layering (averaging ~20° east) is exaggerated. Sample points A–G span the ~200 m section; the plotted mineral compositions — clinopyroxene (En′), olivine (Fo), and plagioclase (An) — trace a differentiation trend that reverses at the "cumulus reversal" arrow. The observation deck is the Thompson Hill rest-area overlook.

> *The cryptic variation of En and Fo across the cumulus regression exposed in this section [. . .] is the reverse of that expect[ed] from magma recharge. [. . .] An alternative explanation to magma recharge is that the textural and compositional variations across this interval reflect decompression of the chamber due to eruption to the surface. Decompression of a volatile-enriched magma would cause supercooling and multiple saturation of the magma and thereby explain the abrupt decreases in grain size and cumulus phase changes without much compositional variation. Magma expulsion through the roof of the layered series would also explain the occurrence of a gabbroic anorthosite inclusion here and elsewhere throughout the cyclic zone.*

This interpretation positions this cumulate as a record of the magma chamber *venting to the surface* — the intrusion and the rift lavas can be thought of as two ends of one plumbing system.

3.2 Stop 3b: an anorthosite xenolith in the anorthositic series (as time allows)

Location: 46.752309°N, 92.175200°W — roadcuts on Skyline Parkway above Oneota Cemetery. This is Stop 2-6 of Miller (2011), taken if the clock allows.

Here Skyline Parkway crosses into the *anorthositic series* of the Duluth Complex (Figure 2.1). This outcrop also has an **anorthosite xenolith**, a lump of nearly pure plagioclase (more than 95%) enclosed in the host gabbroic anorthosite, with a sharp contact against it.

Plagioclase is buoyant within tholeiitic magma at depth, so rather than settling it accumulates as a *crystal mush* at depth, which was then intruded upward into the higher parts of the complex — the anorthositic series being, in Miller's words, an "igneous breccia" built by "repeated intrusions of plagioclase crystal mush" (Miller, 2011). For the xenolith we can consider it to be a block of older, deeper anorthosite torn loose and *entrained and carried upward* within the mush. Standing at a roadcut above Duluth, we are looking at material rafted up from deep in the magmatic system.

3.3 Stop 3c: The anorthositic series (as time allows)

Location: 46.75846°N, 92.16392°W — roadcuts on Skyline Parkway a short way beyond Stop 3b, within Stop 2-6 of Miller (2011). Taken if time allows.

This is a clean exposure in the *anorthositic series*. Rather than name the rock for you, we will leave it as an exercise: estimate the modal proportions of plagioclase, olivine, and pyroxene in the outcrop, plot them on the modal classification diagram (Fig. 2.3), and read off a rock name — anorthosite, gabbroic anorthosite, troctolitic anorthosite, and so on. Where does this rock fall relative to the troctolites and olivine gabbros we named at the DWP Trail?

Onward to rift lavas!

Lakewood — pahoehoe lava flows of the North Shore Volcanic Group

4

At Skyline Parkway we read the cumulus reversal in the layered gabbro as a record of the magma chamber *venting to the surface* — the deep intrusion and the rift lavas as two ends of one plumbing system. Here at Lakewood we standing on the shallowly dipping subaerial lava flows of the North Shore Volcanic Group. Where the Thomson Formation was folded and cleaved and the Duluth Complex crystallized kilometres down, these flows cooled at the surface, were buried beneath the lavas that kept erupting on top of them, tilted gently toward the lake, and have since been exhumed by erosion — a mild history next to that deep plumbing, and mild enough that their ropy tops, their gas cavities, and the very contact between one flow and the next are well preserved. This is Locality 1 of Green (1987), and the description that follows draws on that guide.

Our day closes where the magma did: at the top of the system, on the erupted lava that is the surface expression of everything we have walked through — it would have erupted far-above the deformed Paleoproterozoic footwall at Thomson Dam, the baked basalt and basal contact of the Duluth Complex on the DWP Trail, and the layered and anorthositic series along Skyline Parkway.

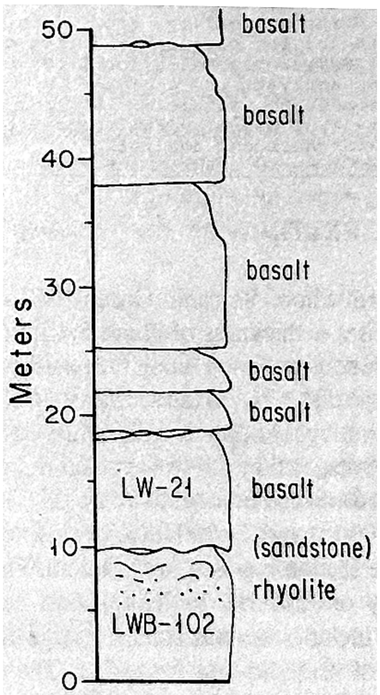

Figure 4.1: The ~50 m volcanostratigraphic succession at the Lakewood shore, from Green (1987) (their Fig. 2B). A thick red rhyolite (sample LWB-102) at the base is overlain by a thin interflow sandstone and then a stack of iron-rich basalt flows (the lowest is sample LW-21). We come in at the rhyolite–basalt contact near the bottom of the column and work up through the flows.

4.1 Location and logistics

Parking: 46.85332°N, 91.97821°W — a paved pull-out on the lake (southeast) side of Scenic North Shore Drive (County 61), about 2 km northeast of the mouth of the Lester River in Duluth (Lakewood 7.5′ quadrangle; NAD 83 UTM zone 15N 577894E, 5189371N).

Route: from the southwest end of the little parking area, a path descends the bluff to the wave-washed outcrop at the lakeshore (about 46.85152°N, 91.98094°W; UTM 577689E, 5189169N); the short walk is drawn in Figure 4.2. The bluff itself is not bedrock: it is red clay deposited in Glacial Lake Duluth as the late Wisconsinan ice sheet melted back from the Lake Superior basin roughly 14,000 years ago — a thin Quaternary veneer draped over lavas a thousand times older.

Time: about one hour along the shore.

Safety
The descent to the shore is down a short but steep path on the bluff — watch your footing and take it slowly. At the water's edge the wave-washed rock is slick, and Lake Superior is cold so be careful.

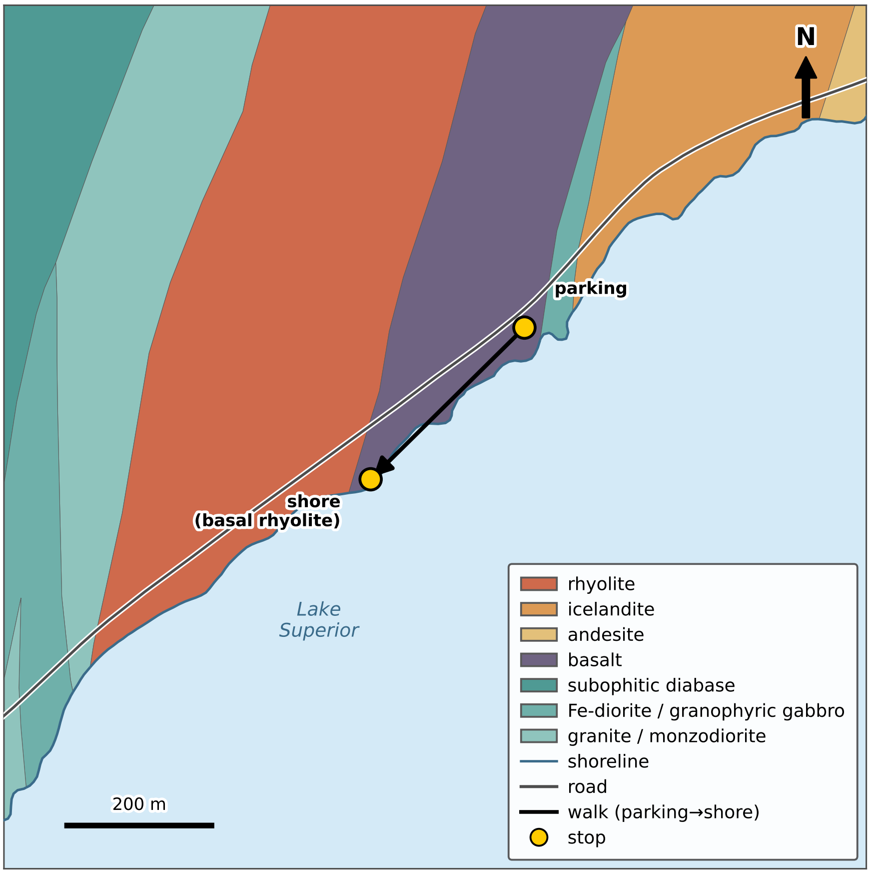

Figure 4.2: Bedrock geologic map of the Lakewood shore stop, with polygons from the Minnesota Geological Survey M-128 mapping of the Lakewood quadrangle (Boerboom et al., 2002). We park on the basalt (46.85332°N, 91.97821°W) and walk southwest down to the shore, where the outcrop is the basal *rhyolite* of the local volcanostratigraphic succession (46.85152°N, 91.98094°W).

4.2 Geologic setting

The North Shore Volcanic Group (NSVG) is the extrusive pile of the Midcontinent Rift volcanics exposed along the Lake Superior shoreline: roughly 9.7 km of dominantly subaerial, tholeiitic flood basalt and basaltic andesite, with subordinate rhyolite and lesser andesite and icelandite, all erupted during the ca. 1.1 Ga rift magmatism (Green, 1982; Green et al., 2011). The succession has the form of half of a broad dish tilted to the southeast, so that near Duluth the flows strike roughly north and dip gently east-southeast, toward the axis of the rift beneath Lake Superior.

This stop sits up from the base of the pile — about 2,800 m above the base of the section exposed near Duluth (Fig. 4.3) — and therefore well *above*, in structural level, the deep plumbing we spent the rest of the day in. The Ely's Peak basalts at the DWP Trail are the earliest, lowest lavas of the same group; the flows here are younger and higher and erupted during the main stage of Midcontinent magmatic activity that the Duluth Complex is also assigned to. The Duluth Complex is where such magma stalled and crystallized slowly, and these flows are where it reached the surface and rapidly cooled upon eruption.

The rhyolite we will see in this sequence is one of the North Shore Volcanic Group's remarkably large felsic units: the group is unusually felsic for a flood-basalt province (~10–25% rhyolite and icelandite), and

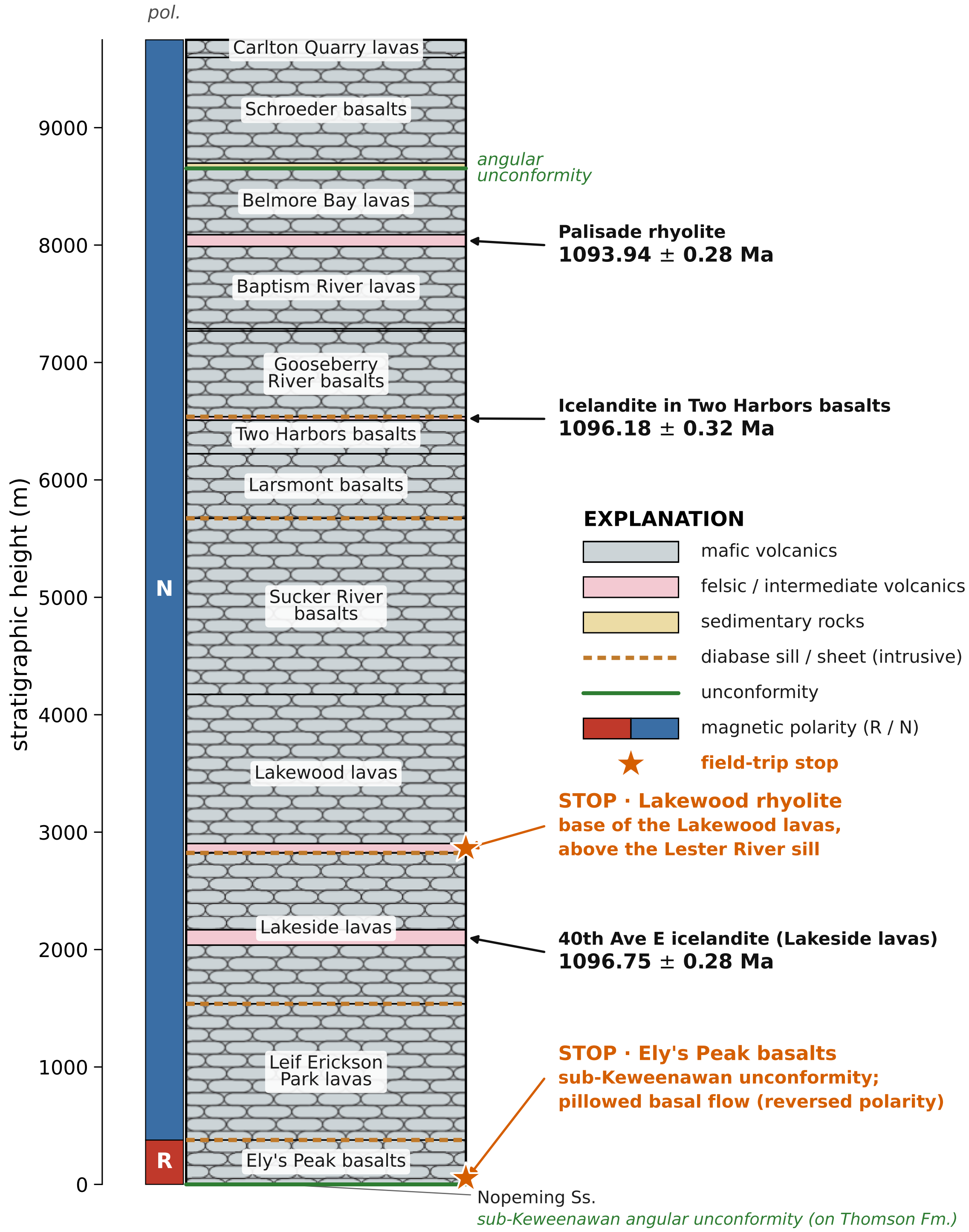

Figure 4.3: Generalized stratigraphy of the southwest (Duluth) limb of the North Shore Volcanic Group, placing the day's two North Shore Volcanic Group stops within the ~9.7 km pile. The Ely's Peak basalts form the reversed-polarity (R) base of the section — the earliest, lowest lavas, visited on the DWP Trail (Chapter 2) — beneath the entire normal-polarity (N) succession above; the Lakewood shore stop is at the Lakewood rhyolite (Green and Fitz III, 1993), the felsic unit at the base of the Lakewood lavas, directly above the Lester River diabase sill. Lithostratigraphic unit names and thicknesses are from Green et al. (2011) (their Table 1), and the diabase sills and sheets of that table are drawn as orange dashed intrusive contacts. The high-precision CA-ID-TIMS ^{206}Pb/^{238}U dates are from Swanson-Hysell et al. (2019); the closest direct age below the stop is the 40th Avenue East icelandite (1096.75 ± 0.28 Ma) in the underlying Lakeside lavas.

Green and Fitz III (1993) documented individual Keweenawan felsic flows with estimated volumes up to ~600 km^3 and lateral extents up to ~40 km — far larger than most silicic lavas. They also argued that many such units are not simple lava flows but *rheoignimbrites*: pyroclastic ash flows erupted so hot and welded so densely that the reconstituted mass then flowed like lava. The only place to tell a rheoignimbrite from a true lava is in the basal decimetres and the very top, where welded shards and fiamme may survive in an otherwise lava-like, crystalline flow — which is exactly the part of the Lakewood rhyolite exposed at the shore. Regionally, the Lakewood rhyolite marks the base of a ~3.5 km sequence of progressively more mafic and primitive flows (Green and Fitz III, 1993), so from this contact the section above youngs *and* grows more primitive upward.

At the shore the key relationship is a *contact between two very different volcanic rocks* (Fig. 4.1). The base of a ~9-m-thick iron-rich (transitional) basalt flow rests directly on the lumpy, flow-banded top of a thick, red rhyolite (Green, 1987); both units appear on the 1:24,000 bedrock map of the Lakewood quadrangle (Boerboom et al., 2002) reproduced in Figure 4.2. Thin lenses and patches of interflow sandstone occur locally along the contact, and calcite fills vesicles and fractures near it. Above the basal flow, a stack of thinner pahoehoe flow units have well-preserved ropy tops. The tight, unbrecciated base, the massive interior with its vesicle cylinders, and the ropy pahoehoe top together record a lava that advanced rapidly over the ground as a fluid sheet.

4.3 Along the shore: what to look for

Features to find

Working along the wave-washed outcrop, see how many of these you can find:

- ☐ **The rhyolite–basalt contact** — the flow-banded top of the red rhyolite, overlain by the base of the basalt flow. Look along it for lenses of interflow sandstone.
- ☐ **Flow banding in the rhyolite** — the streaky, laminar layering that records viscous flow of the stiff, silica-rich lava.
- ☐ **A pahoehoe flow top** — the ropy, coiled surface of a thin flow unit, preserved and exhumed by wave erosion.
- ☐ **Vesicle cylinders** — vertical pipe-like columns of vesicles or amygdules rising through the massive interior of a flow.
- ☐ **Pipe vesicles** — slender vesicles rooted at the *base* of a flow, pointing up into it.
- ☐ **A vesiculation contrast** — an abundantly vesicular / amygdaloidal flow top passing up into the less-vesicular base of the overlying flow.
- ☐ **Amygdule fills** — the secondary minerals filling the vesicles (see the aside below).

The vesicle patterns are neat to observe in these lavas. **Pipe vesicles** at a flow's base form as gas rises out of the chilled, still-liquid sole. **Vesicle cylinders** carry gas and late, differentiated liquid upward through the crystallizing interior. And the **vesicular, amygdaloidal flow top** is where the bulk of the exsolved gas finally froze in place as the flow's upper crust quenched. The sharp drop in vesicularity as you cross from one flow's frothy top into the next flow's tight base marks a *flow contact* — and reading that alternation up the section, as sketched in the stratigraphic column (Fig. 4.1), helps differentiate flows in a volcanostratigraphic succession like this.

A note on the vesicle fillings

The minerals in the amygdules are a common curiosity when looking at these lavas. The North Shore Volcanic Group records a regional, burial-driven progression from zeolite facies up into greenschist facies, and Schmidt and Robinson (1997) mapped it as five mineral zones along the shore from Duluth to Tofte. The Lakewood flows in their Zone 4 — the laumontite–chlorite–albite ± prehnite ± pumpellyite zone — the second-highest grade in the sequence, reflecting the deep structural position of these flows near the base of the pile (Figure 4.3). Peak temperatures here were estimated to be on the order of 220–260 °C.

In this zone, the amygdules are typically laumontite (the calcium zeolite that dominates the amygdaloidal flow tops, in pockets up to a couple of centimetres), along with chlorite, albite, quartz, and calcite, and the diagnostic higher-grade calc-silicates prehnite and pumpellyite. What you won't find are the delicate low-temperature zeolites — thomsonite, scolecite, heulandite, stilbite — that are often the targets for rock hounds and that are found farther northeast toward Tofte, where burial grade was lower. In other words, as we drive up the shore you are heading down the temperature gradient, from prehnite–pumpellyite fillings here to gem thomsonite up north.

4.4 An IRM connection: color, oxidation, and the remanence carriers

These NSVG basalts are overall fantastic recorders of the paleogeography of Laurentia (Swanson-Hysell et al., 2019) and the Mesoproterozoic geomagnetic field (Tauxe and Kodama, 2009). Their remanence in many places is carried by magnetite (and titanomagnetite) that acquired a thermoremanent magnetization (TRM) as each flow cooled. However, this outcrop here gives a direct, visible indication that there are more magnetic minerals to think about.

Where these basalts have been altered by hydrothermal fluids — preferentially along fractures and through the permeable, vesicular flow tops — their primary magnetite has been oxidized to secondary **hematite**, and the rock takes on red and purple hues in place of fresh grey. This hematite should hold *chemical* remanent magnetization (CRM) acquired at the time of that alteration (likely contemporaneous with volcanic activity).

And our time in exposures of the Midcontinent Rift comes to an end. When we are back in Minneapolis, remember that these rocks are under your feet (but buried below Paleozoic sedimentary rocks).

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